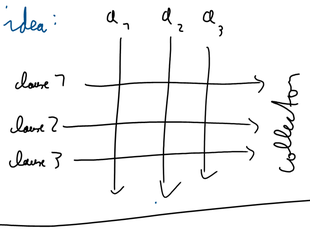
Minesweeper - Component Definition

Samstag, 13. Dezember 2025 17:13

idea:



similar to MCWater

CNF formula satisfiable
⇕
There exists a layout of bombs such that the minesweeper board is valid

Notation: numbers have the same function they have in Minesweeper
all completely empty space is a 0

● ... predefined Bomb

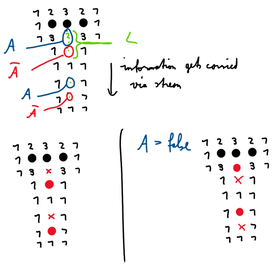
● ... input-dependent Bomb

× ... input-dependent safe field (no Bomb)

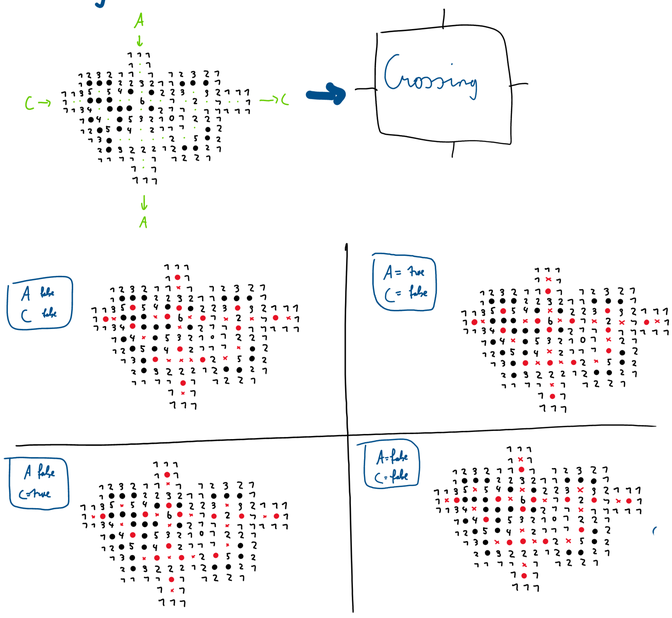
? ... value of variable: true → × , false → ●

· ... field without value, can become × or ● (dependent on input)

Input stream for a variable:



Crossing: clause does not use variable, stream do not change



Close stream

checks variable specified in a clause is 1 if at least one literal satisfied the clause

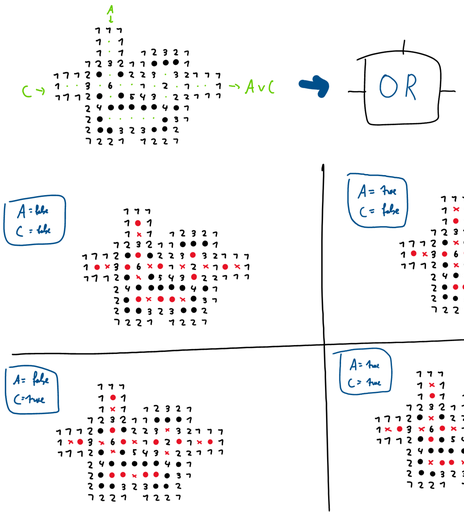
Note: we are "stuck" at one point but by choosing a wrong option → we get an error, so crossing is safe

(did only this, but 7 could be bomb too)

↓ wrong bomb left of 6

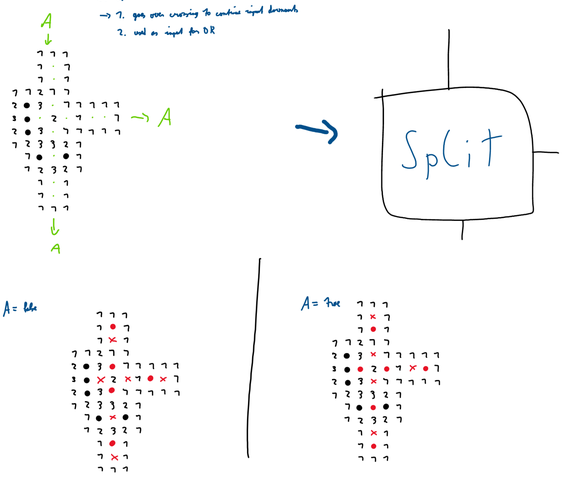
6 has 7 bombs!
NOT VALID

OR: clause is true if clause was already true or if current literal is true



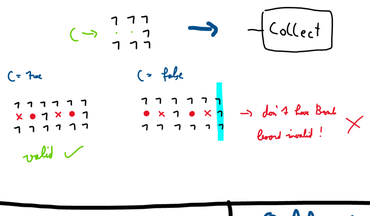
Splitter: Because OR does not keep variable input as secondary output,

we split input from OR to two equal stream
→ 1. goes over crossing to continue input towards
2. out as input for OR

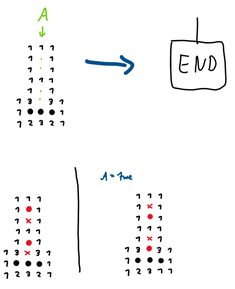


Collector: needs to receive true

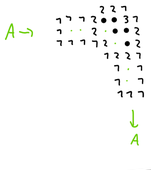
Otherwise clause is false → not satisfying assignment → not valid minesweeper configuration



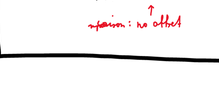
End: ends input stream



Turn stream



Offset stream (horizontally)



Offset stream (vertically)

needed for clause to read negation of variable (when it was negated literal)